
ImageNet-trained CNNs are not biased towards texture: Revisiting feature reliance through controlled suppression

Abstract

The hypothesis that Convolutional Neural Networks (CNNs) are inherently texture-biased has shaped much of the discourse on feature use in deep learning. We revisit this hypothesis by examining limitations in the cue-conflict experiment by Geirhos et al. To address these limitations, we propose a domain-agnostic framework that quantifies feature reliance through systematic suppression of shape, texture, and color cues, avoiding the confounds of forced-choice conflicts. By evaluating humans and neural networks under controlled suppression conditions, we find that CNNs are not inherently texture-biased but predominantly rely on local shape features. Nonetheless, this reliance can be substantially mitigated through modern training strategies or architectures (ConvNeXt, ViTs). We further extend the analysis across computer vision, medical imaging, and remote sensing, revealing that reliance patterns differ systematically: computer vision models prioritize shape, medical imaging models emphasize color, and remote sensing models exhibit a stronger reliance on texture. Code is available at [\[redacted\]](#)

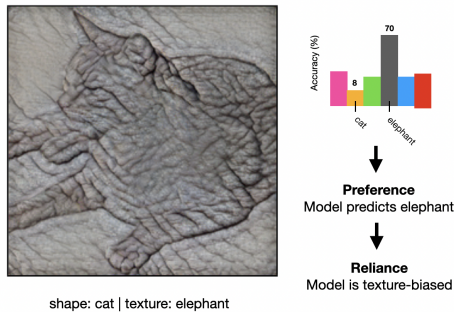
1 Introduction

Convolutional neural networks (CNNs) have played a central role in the development of deep learning models for visual recognition [1], [2], [3], [4]. Their success across a range of computer vision (CV) benchmarks has contributed to the perception that they acquire perceptual representations resembling those of humans [5], [6], [7]. However, a growing body of work suggests that CNNs may process visual information in fundamentally different ways [8], [9], [10]. One of the most influential claims in this direction is that CNNs trained on ImageNet are inherently biased towards texture [8], in contrast to humans who predominantly rely on shape cues [11]. This claim, first formalized by Geirhos et al. [8] through their cue-conflict experiment, has since shaped much of the discourse on how to evaluate and interpret the use of features in deep neural networks.

In the cue-conflict experiment, images are synthesized by combining the shape of one object class with the texture of another, using neural style transfer techniques [12]. Models and humans are then presented with these hybrid images, and their predictions are analyzed to infer which visual cues they rely on. The observed divergence, with CNNs favoring texture and humans favoring shape, has become a dominant narrative for understanding human-machine perceptual differences and has inspired a wide range of follow-up studies [13], [14], [15], [16], [17].

Although influential, the cue-conflict experiment is based on assumptions that may limit the generalizability and clarity of its findings. Conceptually, it reduces feature reliance to a binary choice between shape and texture, overlooking other potentially informative cues such as color, and tends to link salience with reliance implicitly. Methodologically, the generated stimuli entangle unintentionally

Cue-Conflict Setup by Geirhos et al. [8]



Evaluating Feature Reliance through Suppression

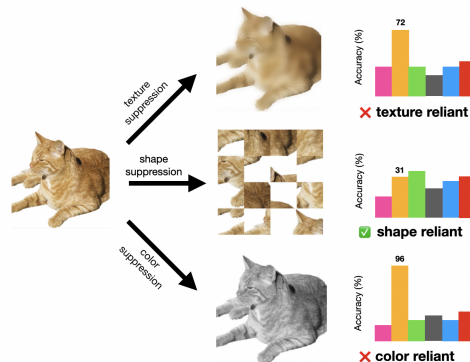


Figure 1: Comparison of cue-conflict setup [8] (left) and our suppression-based framework (right). While Geirhos et al. infer reliance through preference on hybrid images, our framework directly quantifies reliance by measuring accuracy under systematic suppression of texture, shape, or color.

multiple features, introduce texture cues across the image in a spatially unbalanced manner, and rely on shape-based response interfaces that may bias human judgments. As discussed further in Section 3, these conceptual and methodological limitations complicate conclusions about the feature use of models and humans.

In this work, we argue for a conceptual shift: from analyzing feature bias through forced-choice conflicts to assessing feature reliance through targeted suppression. This conceptual distinction reframes how feature preferences and reliance should be evaluated. A model may prefer a certain cue in conflict, not because it is more predictive, but because it is more salient. Conversely, a model may rely heavily on a feature in natural settings, even if it does not dominate in cue-conflict scenarios. To address the aforementioned limitations, we propose a new domain-agnostic evaluation framework that quantifies performance degradation under systematic suppression of individual feature types (e.g., shape, texture, and color), enabling empirical measurement of reliance. The proposed framework does not rely on adversarial inputs or neural style transfer, but instead uses direct feature-suppressing transformations. By isolating individual feature contributions, our framework offers a more reliable basis for interpreting model decisions and comparing representational strategies, both between humans and neural networks, and across model architectures and domains.

Our main contributions are as follows:

- (1) We present a re-examination of Geirhos et al.’s cue-conflict experiment [8], highlighting aspects in their evaluation protocol that may limit its generalizability.
- (2) We introduce a domain-agnostic framework for evaluating feature reliance through targeted feature suppression, enabling cleaner measurement of model dependence on individual visual cues without requiring conflicting cue setups.
- (3) Using the proposed framework, we systematically compare human and model feature reliance under controlled conditions. Our results challenge the texture bias hypothesis [8] by showing that CNNs are not inherently texture-biased; instead, they only exhibit a pronounced sensitivity to local shape, which can be mitigated through modern training strategies. Notably, models trained with vision-language supervision most closely match human behavior.
- (4) We apply the same framework to assess domain-specific differences in feature reliance, showing that models trained on CV, remote sensing (RS), and medical imaging (MI) datasets prioritize distinct visual cues depending on domain characteristics.

2 Related Work

Understanding which features deep neural networks rely on for image classification has been a long-standing research question. While early interpretations of CNNs assumed a hierarchical buildup from

low-level edges to complex shape representations [5], [6], [7] more recent studies have challenged this view, suggesting that CNNs often rely disproportionately on local texture rather than global shape [9], [10], [8], [18]. Geirhos et al. [8] formalized this observation as the texture bias hypothesis, using a cue-conflict protocol to reveal divergent feature preferences between humans and CNNs.

Subsequent work investigated factors shaping feature reliance beyond architecture. Hermann et al. [13] showed that texture bias in CNNs arises primarily from training objectives and augmentations, with techniques like blurring and cropping increasing shape bias more than architectural changes. Although shape features are present in deeper layers [15], [14], they are not consistently used during classification. Transformer-based models and vision-language models have shifted this discussion. Vision transformers (ViTs) exhibit lower texture bias due to their global attention mechanism [19], [20], and vision-language models show improved alignment with human-like shape use [21].

Various methods have attempted to enforce shape bias or suppress texture cues for improved robustness, including anisotropic filtering [22], edge encoding [23], style disentanglement [24], [25], and shape-focused augmentations [26], [27]. However, stylization alone may improve robustness independent of shape bias [28], and neither shape nor texture bias reliably predicts generalization [16]. These findings have motivated integrative approaches that combine diverse feature biases. Joint supervision [29], ensembles [30], and adaptive recombination [31] aim to harness complementary features. Ge et al. [32] and Jain et al. [17] show that disentangling and combining shape, texture, and color improve robustness and interpretability. Nonetheless, Lucieri et al. [33] caution that in domains like MI, cue entanglement is essential and biasing towards shape may be counterproductive.

Efforts to increase shape bias are often motivated by the broader goal of human-model alignment. Geirhos et al. [34], [35] show that even robust models exhibit error patterns that diverge from humans, revealing a persistent consistency gap. Muttenthaler et al. [36] further argue that alignment with human conceptual structure depends more on training signals than model scale, indicating that robustness and shape bias alone are insufficient proxies for human-like perception.

3 Rethinking Texture Bias: A Critical Look at Cue-Conflict Evaluation

The hypothesis that CNNs trained on ImageNet are biased towards texture was popularized by Geirhos et al. [8], who introduced a cue-conflict evaluation protocol. In this protocol, images were generated by neural style transfer [12], combining the shape content (cue) of one class with the texture content (cue) of another. Predictions from both humans and CNNs on these images were then used to infer whether classification decisions were driven more by shape or texture features. Over time, the cue-conflict evaluation protocol has become a de facto standard for assessing feature bias in deep neural networks. While impactful, this protocol introduced several assumptions and limitations that have received limited attention. Conceptually, the protocol frames feature reliance as a binary shape-or-texture choice, which may overlook other cues such as color and conflates preference with dependence. In addition, the stylized stimuli constrain the evaluation of feature bias to naturalistic images with a similar set of classes and cannot be generalized across datasets (e.g., flower classification) or domains (e.g., RS, MI). Beyond these conceptual limitations, the cue-conflict protocol exhibits three methodological concerns in its design and implementation:

- (i) **Lack of Feature Isolation.** The texture cues within the cue-conflict images also preserved information beyond texture, including color and local shape structures (e.g., contours and parts of silhouettes). As a result, the synthesized texture cue was not a pure representation of texture but a composite of multiple features, making it difficult to attribute classification behavior to texture alone. An example can be seen in Figure 2a.
- (ii) **Overloaded Texture Class Signals.** The protocol consistently inserted texture cues not only into the object region but also into the image background. Since CNNs aggregate local statistics across spatial positions, this broad spatial distribution increases the signal strength of the texture class relative to the shape class. This spatial imbalance systematically biases CNNs towards texture-based decisions, not because of an intrinsic preference but due to the dominant spatial availability of the texture signal. An example can be seen in Figure 2b.
- (iii) **Human Interface Bias Towards Shape.** Participants in the human experiments selected the image class by clicking on buttons labeled with icons representing each category. These icons represented global shape characteristics (i.e., silhouettes), potentially guiding participants

towards matching shape features in the cue-conflict image with the icon. This response format potentially introduces bias towards shape decisions, especially when participants were unsure which feature to prioritize. The used icons are visualized in Figure 2c.

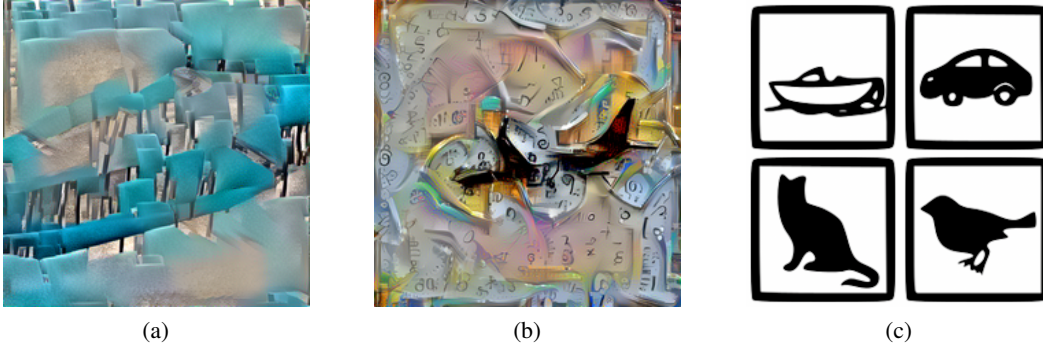


Figure 2: Example images taken from the cue-conflict dataset [8]. **(a)** Boat shape cue merged with chair texture cue. **(b)** Airplane shape cue merged with clock texture cue. **(c)** Icons of the human interface to select classes.

These design choices may inadvertently influence CNNs towards texture-driven decisions and humans towards shape-based decisions and complicate drawing definitive conclusions about the actual feature reliance of models.

4 A Domain-Agnostic Framework for Feature Reliance

Accurately assessing how deep neural networks rely on different visual features remains a central challenge in understanding their behavior. While the cue-conflict evaluation protocol [8] introduced a reliance test based on feature bias, it imposes conceptual and methodological constraints that limit its generalizability. Rather than forcing models to choose between shape and texture, we propose to assess their reliance on individual feature types by systematically suppressing them and measuring the resulting impact on classification performance. This shift enables a more flexible, generalizable, and semantically grounded analysis of feature use in neural networks.

To evaluate the reliance of deep neural networks on individual visual features, we employ a set of image transformations that selectively suppress shape, texture, or color information while minimally affecting the remaining features. Each transformation is chosen for its ability to target a specific feature class. We define three feature types:

- **Shape** refers to information carried by spatial arrangement and structural contours, including both global (object outline) and local (part-level) shape.
- **Texture** is defined by repetitive patterns, high-frequency local variations, and fine-grained surface details.
- **Color** denotes chromatic information independent of spatial layout or texture.

For each feature type, we include two complementary transformations that differ in their suppression mechanisms and preservation profiles, offering distinct but comparable perspectives on the targeted feature. The transformations are summarized in Table 1 and briefly described in the following. Patch Shuffle [37], [28] and Patch [redacted] disrupt shape by modifying non-overlapping image patches: Shuffle randomizes spatial positions, while [redacted] preserves locality of patches but breaks edge continuity. Both affect global or local shape, depending on the grid size. Bilateral Filtering [38] and Gaussian Blur reduce texture by smoothing high-frequency details, with the former preserving edges more effectively. Grayscale removes chromatic cues entirely, while Channel Shuffle disrupts color correlations without altering intensity. In the following, we validate the suppression effects of these transformations using quantitative metrics.

Table 1: Feature suppression transformations used in this work. Each feature is suppressed using two transformations with differing strengths.

Feature Type	Transformation 1	Transformation 2
Shape	Patch Shuffle	Patch  tion
Texture	Bilateral Filter	Gaussian Blur
Color	Grayscale	Channel Shuffle

Table 2: Quantitative validation of suppression transformations across 800 images of ImageNet. Each transformation is used with a fixed parameter setting (see Param ID legend below). Values report normalized metric scores. Arrows indicate desired direction: $\uparrow$ higher is better, $\downarrow$ lower is better.

Transformation	Param ID	Texture $\downarrow$	Shape $\uparrow$	LV $\downarrow$	HFE $\downarrow$	ESSIM $\uparrow$	GC $\uparrow$
<i>Texture-Suppressing</i>							
Bilateral Filter	A	0.521	0.796	0.548	0.493	0.737	0.855
Box Blur	B	0.193	0.363	0.237	0.148	0.436	0.289
Gaussian Blur	C	0.349	0.662	0.392	0.306	0.744	0.579
Median Filter	D	0.357	0.506	0.399	0.316	0.584	0.429
NLMeans Denoising	E	0.706	0.797	0.723	0.690	0.730	0.864
Transformation	Param ID	Texture $\uparrow$	Shape $\downarrow$	LV $\uparrow$	HFE $\uparrow$	ESSIM $\downarrow$	GC $\downarrow$
<i>Shape-Suppressing</i>							
Patch Shuffle	F	1.000	0.176	1.000	1.000	0.205	0.147
Patch  tion	F	1.000	0.293	1.000	1.000	0.339	0.247

Legend: A: $d=11$, $\sigma_c=170$, $\sigma_s=75$; B: $k=11$; C: $k=11$, $\sigma=2.0$; D: $k=11$; E: $h=20$, $tw=11$, $sw=11$; F: grid=6.

4.1 Quantitative Validation of Suppression Transformations

While the individual transformations used in this work are not novel, their selection for targeted feature suppression requires empirical justification. To validate that each transformation suppresses the intended visual feature (e.g., texture, shape) while preserving others, we quantify their effects using four metrics: Local Variance (LV) [39] and High-Frequency Energy (HFE) [40] to assess texture suppression, and Edge-SSIM (ESSIM) [41] and Gradient Correlation (GC) to measure shape preservation. All metrics are normalized to the range $[0, 1]$ by dividing by the scores of the unsuppressed (i.e., original) image. Higher values of ESSIM and GC indicate better preservation of edge and structural information, while lower values of LV and HFE reflect stronger suppression of texture features. Further, we compute a harmonic mean across the two texture metrics (Texture) and the two shape metrics (Shape) for each transformation.

We test the effectiveness of the feature suppression transformations across 800 sampled images from the ImageNet validation set. For each transformation, we evaluate a representative parameter setting chosen to balance suppression of the target feature and preservation of others. The respective parameters, such as kernel size or smoothing strength, are indexed by Param IDs in Table 2, with details listed below the table. A full ablation of different parameter settings is provided in the supplemental material (see Section D). In addition to our selected texture suppression transformations, we also compare common alternatives such as Non-Local Means Denoising [42], Box blur, and Median filtering [43] to ensure a fair comparison across standard smoothing techniques. Among texture-suppressing methods, bilateral filtering yields the most balanced trade-off between reducing texture (LV: 0.54, HFE: 0.49) and preserving shape (ESSIM: 0.74, GC: 0.85). Gaussian Blur suppresses texture more uniformly but leads to a greater loss of shape information. Box blur and median filtering remove texture strongly, but at a substantial cost to shape preservation. For shape suppression, we evaluate Patch Shuffle and Patch  tion with a grid size of 6. These transformations preserve texture but substantially disrupt structural contours, making them suitable for assessing shape reliance. To complement the quantitative evaluation, qualitative visual examples of the suppression effects are provided in the supplemental material (see Section C).

5 Experiments

5.1 Experiment I: Human vs. CNNs Feature Reliance

Experimental Setup. To compare human and model reliance on different visual features, we designed a controlled experiment inspired by Geirhos et al. [34], [8]. We constructed an ImageNet16-like dataset by selecting 50 representative images for each of 16 entry-level categories derived from the WordNet hierarchy [44] (see [34] for details). Images were selected based on the most confidently predicted samples in the ImageNet validation set [45] by a ResNet50 [2] pretrained on ImageNet1k, ensuring balanced subclass coverage. For categories with insufficient confident predictions (airplane, knife, oven), additional samples were manually added. All images were resized to 224×224 pixels.

Humans were presented with image stimuli in randomized order under one of five conditions: original, global shape suppression, local shape suppression, texture suppression, or color suppression. Each feature was suppressed via a single transformation with fixed hyperparameters: Patch Shuffle with grid size 3 (global shape), grid size 6 (local shape), bilateral filtering with $d=12$, $\sigma_{\text{color}}=170$ and $\sigma_{\text{space}}=75$ (texture), and grayscale conversion (color). See Section 4.1 for justification. Each participant saw only one randomly chosen version of each image to avoid learning effects. The five suppression conditions of one image were split across groups of five participants to ensure balanced coverage. Twenty participants completed the study. Following Geirhos et al. [8], each trial included a 300 ms fixation square, 200 ms image presentation, and 200 ms pink noise mask (1/f spectral shape) to minimize feedback processing. Participants selected one of 16 categories via a 4×4 grid of alphabetically sorted class names. An additional “not clear” button was available for unrecognizable stimuli. Attention checks were administered every 100 trials, and failed trials were excluded. Additional details and interface screenshots can be found in the supplemental material.

Model evaluation mirrored the human protocol, evaluating their performance under the same five suppression conditions using the identical image set shown to humans. For each image, the class prediction was computed by summing softmax outputs over all ImageNet subclasses mapping to the same entry-level category. Only predictions above the threshold of 0.5 were considered correct. This procedure was chosen heuristically, complementary results using argmax to define class predictions are reported in the supplemental material and show nearly identical reliance profiles.

We evaluated several architectures: ResNet50-standard, trained from scratch with basic augmentations, and ResNet50-sota, trained with a modern recipe [46]. Additional CNNs include MobileNetV3 [47], EfficientNet [3], EfficientNetV2 [48], ConvMixer [49], ConvNeXt [4], and ConvNeXtV2 [50]. Transformer-based models include ViT [51], DeiT [52], SwinTransformer [53], and CLIP ViT [54]. All models except ResNet50-standard were obtained as pretrained checkpoints from the `timm` library [55]. The detailed training procedures can be found in the supplemental material.

Results. Figure 3 presents a comparative overview of the performance of humans and CNNs under feature suppression, plotted as the relative accuracy (i.e., accuracy under suppression divided by baseline accuracy on original images). Separate subplots show results for each suppressed feature type. We highlight three representative CNNs: ResNet50-standard, ResNet50-sota, and ConvNeXtV2 alongside human performance. The results show that CNNs are not strongly reliant on texture: under texture suppression, ResNet50-standard retains 80% of its original performance, close to performance under global shape suppression (83%). The highest vulnerability is observed under local shape suppression, where accuracy drops to just 28%. Humans exhibit a similar reliance profile with local shape suppression being most disruptive, but show higher robustness to it (76% retained accuracy). Interestingly, modern training strategies substantially mitigate this effect: the ResNet50-sota reaches 62% under local shape suppression, and ConvNeXtV2 improves further to 65%. These results suggest that the heavy reliance on local shape observed in earlier CNNs is not architectural in nature but can be alleviated through better training regimes. A likely contributing factor is the inclusion of stronger regularization, improved data augmentations, and more extensive training schedules in the modern setup, which may encourage broader feature utilization beyond local patterns. Statistical significance tests confirming these differences are reported in the supplemental material.

Broadening the analysis to a wider range of architectures (Table 3), we observe that several models trained with state-of-the-art recipes exhibit a more balanced reliance profile. However, this trend is not universal: ConvMixer, EfficientNet, and MobileNet variants retain a strong dependence on local shape, indicating that improved training alone does not guarantee human-like feature use and that architectural inductive biases or capacity limitations may still play a role. Among transformer-

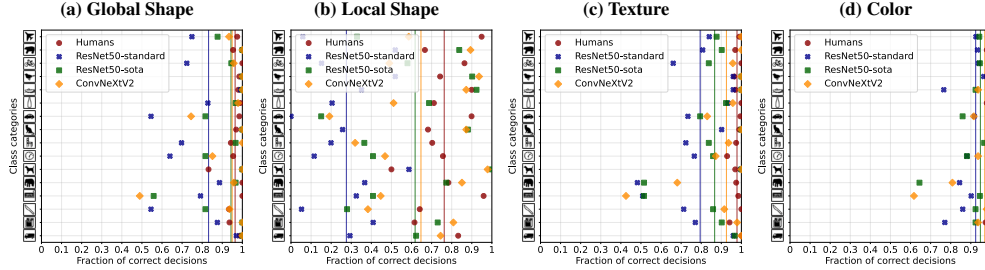


Figure 3: Relative accuracy under feature suppression for human observers and three CNNs ResNet50-standard, ResNet50-sota, ConvNeXtV2 on the curated ImageNet16 dataset. Each subplot shows performance under suppression of a specific feature: **(a)** global shape via Patch Shuffle (grid=3); **(b)** local shape via Patch Shuffle (grid=6); **(c)** texture via bilateral filtering; and **(d)** color via grayscale.

Table 3: Relative accuracy (accuracy under suppression divided by accuracy on original images) for each feature suppression type across models and human observers.

Architecture	Global Shape	Local Shape	Texture	Color	Original	#Params
Humans	0.965	0.763	0.979	0.999	0.969	–
ResNet50-standard [2]	0.832	0.276	0.795	0.924	0.954	25.6M
ResNet50-sota [46]	0.943	0.618	0.867	0.948	0.931	25.6M
ConvNeXt [4]	0.938	0.606	0.910	0.961	0.934	28.6M
ConvNeXtV2 [50]	0.949	0.647	0.925	0.969	0.940	28.6M
EfficientNet [3]	0.870	0.240	0.892	0.987	0.856	30.0M
EfficientNetV2 [48]	0.926	0.423	0.897	0.957	0.932	24.0M
MobileNetV3 [47]	0.795	0.217	0.761	0.859	0.881	5.4M
ConvMixer [49]	0.920	0.437	0.815	0.891	0.874	21.1M
ViT [51]	0.930	0.636	0.921	0.977	0.929	86.6M
DeiT [52]	0.938	0.730	0.926	0.969	0.932	86.6M
Swin [53]	0.924	0.713	0.906	0.941	0.945	87.8M
CLIP ViT [54]	0.959	0.758	0.949	0.984	0.936	86.6M

based models, the ViT demonstrates a feature reliance profile similar to ResNet50-sota across all suppression conditions, challenging the notion that transformers are inherently more shape-oriented than CNNs. Notably, the CLIP ViT model most closely matches human performance across all feature suppression conditions, suggesting that vision-language supervision encourages more human-aligned representations. This may reflect the effect of contrastive vision-language training, which prioritizes alignment with high-level semantic concepts over low-level visual cues.

These findings challenge the texture bias hypothesis popularized by Geirhos et al. [8] as a fixed inductive bias of CNNs. Instead, the observed behavior in the cue-conflict experiment may have reflected a dominant reliance on local shape features, rather than an inherent texture bias.

5.2 Experiment II: Domain-specific Feature Reliance

While Section 5.1 focuses on comparing feature reliance between humans and CNNs on a fixed benchmark, this section explores how reliance on shape, texture, and color varies across domains. The same suppression-based framework introduced earlier is applied to three representative visual domains: CV, MI, and RS. In each case, we fix the architecture to a ResNet50 and apply the standard training protocol, including only the data augmentation techniques random resized crop and horizontal flip. For CV datasets, we either train from scratch or initialize models with ImageNet-pretrained weights (standard training protocol) and then fine-tune on the respective datasets. For MI and RS, we train from scratch to allow a disentangled comparison across domains. Additional results for MI and RS with pretrained models to simulate operational scenarios can be found in the supplemental material. Details about the hyperparameter, as well as an overview of the corresponding validation accuracies, are provided in the supplemental material. In contrast to the previous experiment, in this

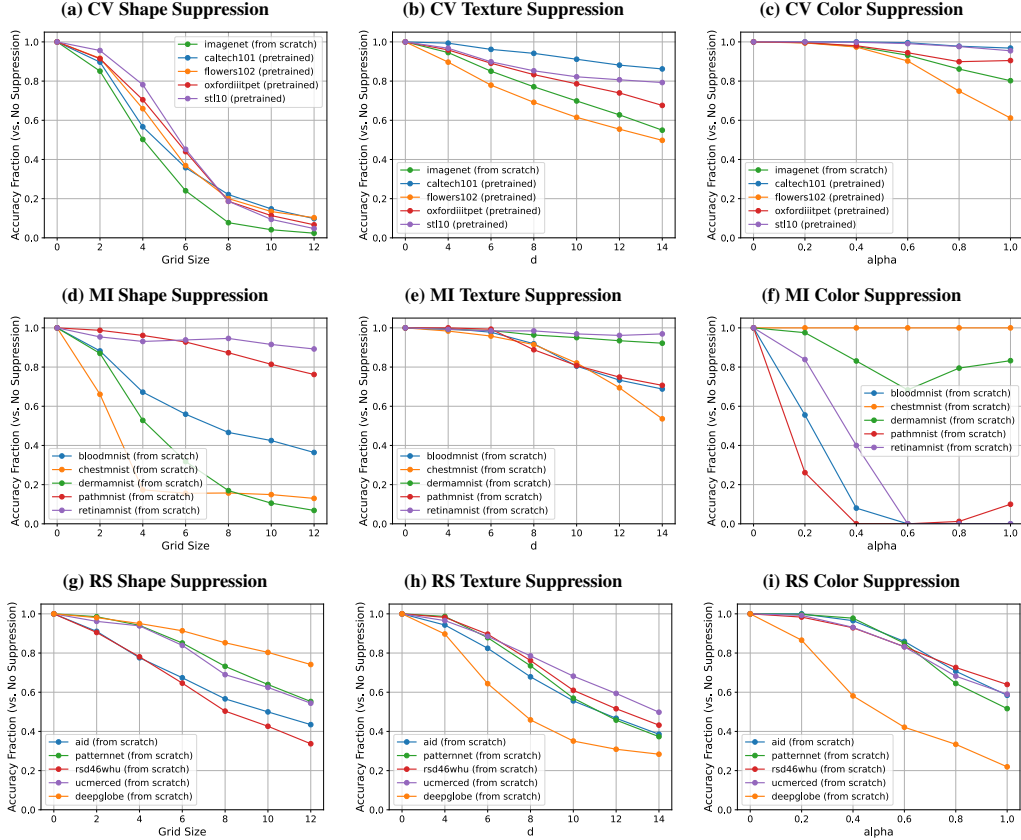


Figure 4: Feature suppression results across three domains. **Top row (a–c):** ResNet50 pretrained on ImageNet and fine-tuned on CV datasets. **Middle row (d–f):** ResNet50 trained from scratch on MI datasets from MedMNIST-v2. **Bottom row (g–i):** ResNet50 trained from scratch on high-resolution RS datasets. Columns correspond to (a, d, g) shape suppression (Patch Shuffle), (b, e, h) texture suppression (Bilateral Filter), and (c, f, i) color suppression (Grayscale).

experiment, suppression strength is treated as a continuous hyperparameter and systematically varied to obtain suppression curves that characterize feature reliance across domains. To reduce redundancy, we report results using one representative suppression technique per feature type in the main paper. Results using alternative suppression methods per feature type are included in the supplemental material and exhibit qualitatively similar patterns across domains.

To visualize domain-specific suppression sensitivity, we present a composite figure of per-domain results in Figure 4, showing the effect of suppressing shape, texture, and color for datasets from each domain. To ensure comparability across datasets with different numbers of classes and baseline accuracies, we standardize performance by rescaling: chance-level accuracy is mapped to 0, and baseline accuracy (i.e., accuracy on original images) is mapped to 1. Relative accuracy under suppression is then expressed on this normalized scale, facilitating direct comparison of feature reliance across domains and datasets. Finally, to synthesize the findings, we aggregate suppression curves in a domain-level comparison (Figure 5) by averaging results across datasets within each domain.

Computer Vision (CV). Figure 4a–c shows suppression results for five standard CV benchmarks (ImageNet [45], Caltech101 [56], Flowers102 [57], Oxford-IIIT-Pet [58], STL10 [59]). Across datasets, we observe that shape suppression induces the strongest performance degradation, especially as the patch shuffle grid size increases. This confirms a pronounced reliance on local shape information in pretrained CNNs. In contrast, texture suppression via bilateral filtering has minimal effect, and color suppression through grayscale conversion yields only minor degradation, indicating that CNNs fine-tuned on these datasets are largely robust to the removal of texture and color cues. These results

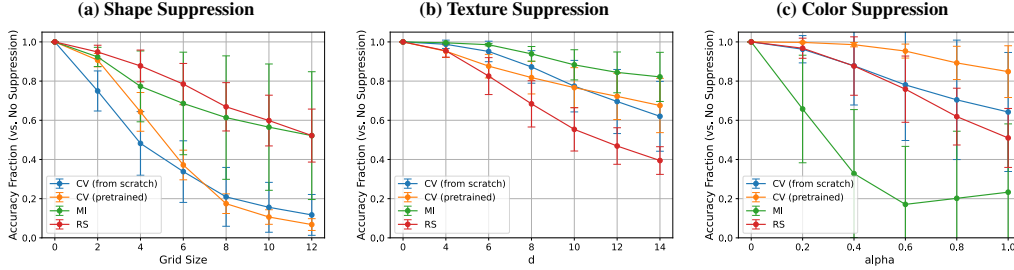


Figure 5: Domain-averaged feature suppression curves for CV, MI, and RS. **(a)** Shape suppression via Patch Shuffle. **(b)** Texture suppression via bilateral filtering. **(c)** Color suppression via grayscale.

are consistent with our human comparison study and suggest that local shape continues to dominate feature reliance in natural image classification tasks. For completeness, the supplemental material includes results for models trained from scratch as well as a class-wise analysis for ImageNet, which confirms that the global reliance patterns are consistent across categories.

Medical Imaging (MI). Figure 4d–f summarizes results on five datasets from the MedMNIST-v2 collection [60]: PathMNIST, RetinaMNIST, BloodMNIST, DermaMNIST, and ChestMNIST. We use the standardized 224×224 pixels version to ensure consistency with the experimental setup. Across these datasets, suppression effects are more heterogeneous than in CV. While shape suppression degrades performance, the impact is generally less pronounced, and texture suppression yields moderate performance drops in datasets such as PathMNIST and BloodMNIST, but relatively little effect in RetinaMNIST and DermaMNIST. By contrast, color suppression induces a substantial decline in classification accuracy for most datasets, reflecting the strong diagnostic role of chromatic cues, except in ChestMNIST, which contains only grayscale images. Taken together, these results suggest that feature reliance in MI varies substantially across datasets, with a common trend towards greater dependence on color information.

Remote Sensing (RS). Figure 4g–i reports suppression curves for five very-high-resolution RGB datasets: UCMerced [61], RSD46-WHU [62], DeepGlobe [63], PatternNet [64], and AID [65]. As in MI, shape suppression impacts performance, but the degradation is less pronounced than in the CV domain, indicating lower reliance on local shape. In contrast to CV and MI, texture suppression leads to substantial performance degradation across all datasets, suggesting that fine-grained surface patterns are critical for RS classification. Surprisingly, color suppression also results in notable performance drops, despite the use of RGB imagery only. This likely reflects strong correlations between chromatic cues and semantic land cover categories. Overall, RS models exhibit a pronounced reliance on texture and color, and comparatively less dependence on local shape, reflecting the distinct statistical structure and spatial semantics of RS imagery.

Cross-Domain Comparison. To synthesize these observations, Figure 5 presents the domain-averaged suppression curves for each feature, including 1-sigma error bars. Three clear trends emerge. First, CV models are most reliant on local shape, especially when trained from scratch, while ImageNet pretraining induces slightly greater robustness. Second, MI models exhibit stronger dependence on color, consistent with the nature of some medical tasks (e.g., in dermatology, histopathology), which often require interpreting chromatic cues. Third, RS models exhibit the highest texture reliance among the three tested domains. This may reflect the nature of many RS classes that are defined by texture-like patterns (e.g., fields, residential areas), rather than by distinct global contours. These patterns confirm that feature reliance is shaped not only by architecture and training regime, but also by the visual and semantic properties of the task or domain.

Finally, to validate the observed feature reliance patterns, we conduct complementary experiments on CV datasets with simultaneous suppression of two features (see Section I.5 in the supplemental material). Results confirm the trends of single-feature suppression: performance is highest when only shape is preserved, reduced when only texture remains, and nearly lost when only color is available. In summary, the findings highlight that domain characteristics, alongside architecture and training regime, play a crucial role in shaping feature reliance. While prior work emphasized architecture-induced biases, our results suggest that data properties equally govern the perceptual strategies that models adopt.

6 Conclusion

This paper revisited the widely cited claim that CNNs trained on ImageNet are inherently biased towards texture. We identify critical conceptual and methodological limitations in the cue-conflict experiment popularized by Geirhos et al. [8] that support this hypothesis. Further, we propose a new framework for evaluating feature reliance based on targeted suppression rather than forced-choice preference. Using this framework, we find no evidence for an inherent texture bias in CNNs, but instead observed a pronounced reliance on local shape features. Nonetheless, we show that this reliance can be substantially mitigated through modern training strategies. Across domains, we find that feature reliance varies substantially: CV models prioritize shape, MI models rely more evenly on color, and RS models exhibit strong texture sensitivity. These findings challenge the notion of fixed architectural biases and instead position feature reliance as a flexible property shaped by optimization objectives and domain-specific semantics, offering new directions for designing models that better align with human perceptual strategies. At the same time, the relative contributions of architectural components and training strategies to these reliance patterns remain to be systematically evaluated.

Limitations. Our framework relies on operational definitions of shape, texture, and color based on specific transformations, but features are continuous and interdependent, limiting perfect isolation. In practice, suppression only reduces rather than eliminates features: texture suppression can leave residual low-level features perceptible as texture, while shape suppression does not fully remove all shape cues. This reflects the inherent trade-off of reducing one feature while preserving others, making absolute removal unattainable. The applied suppression techniques may also introduce artifacts that affect model behavior independently of the targeted features (e.g., block-like structures from Patch Shuffle, smoothing from filtering). Further, the results obtained with pretrained models may reflect effects of similarities between suppression transformations and augmentation techniques (e.g., Cutout and Patch Shuffle). Finally, our human experiments employed a controlled forced-choice design with brief exposures and a limited set of categories to ensure comparability. While necessary for experimental control, these constraints may not fully reflect the richness and adaptability of human visual perception in real-world settings.

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